

System Overview

HelpHive is a production-grade, real-time Progressive Web Application (PWA) built to connect individuals needing everyday assistance (Posters) with nearby verified helpers (Taskers). The platform is engineered for sub-second spatial matching, dynamic wave dispatching, and high-concurrency event broadcasting.

Core Technology Layers

Frontend & User Interface

CLIENT LAYER

- React 19:** Declarative component architecture & modern hooks
- Vite 8:** Fast module bundling & HMR
- TailwindCSS v4:** Utility-first styling & design tokens
- Lucide React:** Vector iconography system
- React Context API:** Global state management (`AppContext` , `NotificationContext`)
- Ola Maps API:** Interactive maps, routing & geocoding

Backend & Database (BaaS)

DATABASE LAYER

- Supabase BaaS:** Full backend infrastructure suite
- PostgreSQL DB:** Enterprise relational database engine
- GiST Spatial Indexing:** PostGIS spatial index for proximity matching
- Row Level Security (RLS):** Fine-grained table security policies
- Supabase Auth:** Mobile-friendly PKCE authentication
- Supabase Realtime:** Postgres Change Broadcasts & RPC Queues

PWA & Serverless Edge

EDGE LAYER

- Vite PWA Plugin:** Web App Manifest & asset generator
- Service Worker (`sw.js`):** Offline shell caching & push handler
- Supabase Edge Functions:** Deno serverless runtime (`push-notification`)
- W3C Web Push API:** Native device push notifications via VAPID keys

Testing, Quality & DevOps

TOOLING

- Playwright:** Multi-role E2E automated testing framework
- ESLint v10:** Modern flat configuration with hooks rules
- Husky v9 & lint-staged:** Pre-commit git hooks
- GitHub Actions:** Automated CI/CD build & test pipeline
- Vercel:** Global edge hosting deployment

Key Package Dependencies

PACKAGE NAME	VERSION	TYPE	ROLE / CAPABILITY
react / react-dom	^19.2.6	Dependency	UI Framework & Rendering Engine
@supabase/supabase-js	^2.106.2	Dependency	Supabase Client SDK (Auth, DB, Realtime, RPC)
lucide-react	^1.16.0	Dependency	Vector Icon Library
vite	^8.0.12	DevDependency	Frontend Build Tool & Dev Server
tailwindcss	^4.3.0	DevDependency	Utility-First CSS Engine
vite-plugin-pwa	^1.3.0	DevDependency	PWA Service Worker & Manifest Builder
playwright	^1.61.1	DevDependency	End-to-End Automated Testing
eslint	^10.3.0	DevDependency	Static Code Analysis & Linting

Repository Structure Map

HelpHive/	
├─ .github/workflows/ci.yml	# Automated CI/CD Pipeline
├─ public/	# Web manifest, icons & public PWA assets
├─ src/	
│ ├─ components/	# Reusable UI elements (MapView, JobCard)
│ ├─ config/	# Supabase client & environment constants
│ ├─ hooks/	# Custom React hooks (useProfileCompletion)
│ ├─ screens/	
│ │ ├─ poster/	# Job creation, live tracking, ratings
│ │ └─ tasker/	# Tasker dispatch feed, job details, onboarding
│ ├─ services/	# Supabase RPC & API service layers
│ ├─ store/	# React Context providers (App, Toast, Notify)
│ └─ sw.js	# Custom PWA Service Worker (Push Handler)
├─ supabase/	
│ ├─ functions/	# Deno Edge Functions (push notifications)
│ └─ migrations/	# PostgreSQL schema migrations & RLS policies
└─ vite.config.js	# Vite configuration & PWA setup